Class 28: Reciprocal Space – 2: Condition for Diffraction

We are proceeding with a general discussion of how waves interact with matter. Later we will look at how electron waves interact with the periodic structure of the same material in which the electrons exist. In our earlier discussions, we noted that the wave vector $k = \frac{2\pi}{\lambda}$, has reciprocal length dimensions, and therefore belongs to 'reciprocal space'. It is therefore of interest to understand how waves interact with a periodic structure, depicted using a reciprocal lattice in reciprocal space.

We found that the reciprocal lattice vector $\overrightarrow{H_{hkl}}$ is perpendicular to the plane (hkl), and that $d_{hkl} = \frac{1}{|\overrightarrow{H_{hkl}}|}$

Let us now look at diffraction as it is represented in real space and as it is represented in reciprocal space.

Figure 28.1 below shows how a parallel set of monochromatic waves, which are in phase with each other, interact with planes of atoms. It is important to note that 'planes of atoms' is itself a concept we have created for our convenience. In reality, atoms in a crystalline material sit at specific locations, or can be thought of as having organized addresses associated with each of them. When viewed along specific directions in space, several atoms appear to reside on specific planes. These planes are then described as planes of atoms. The atoms represent discrete points on the plane and hence the planes are not 'solid' planes in that sense.

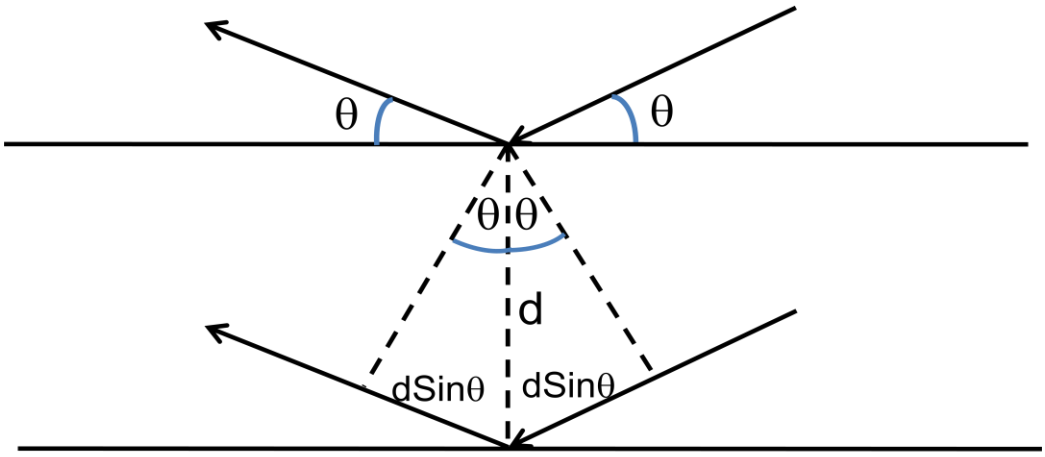


Figure 28.1: Interaction of parallel monochromatic waves, which are in phase, with planes of atoms in a crystalline material

As seen from Figure 28.1, the additional path travelled by the second ray is

$$d \sin\theta + d \sin\theta = 2d \sin\theta$$

When this additional distance travelled by the second ray is an integral multiple of the wavelength λ, constructive interference occurs and strong diffraction peaks are observed.

Therefore the condition for diffraction in real space is:

$$n\lambda = 2d \sin\theta$$

Which is the Bragg law for diffraction.

Let us now examine, how the condition for diffraction may be indicated using reciprocal space notation. Consider the interaction of two parallel rays with two atoms, one located at the origin 'O', and one at a lattice point 'A', as shown in Figure 28.2 below.

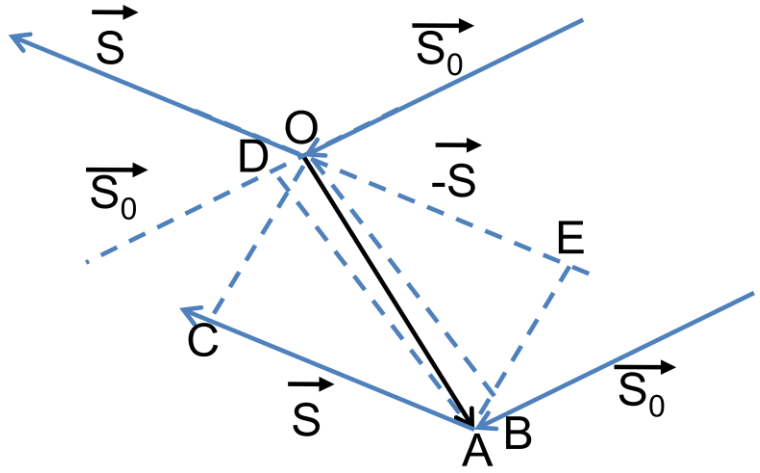


Figure 28.2: The interaction of parallel rays with atoms at two lattice points. One at the origin 'O', and one at a designated lattice point 'A'

Since $\overrightarrow{OA}$ is a valid real lattice vector, since 'A' is a designated lattice point, we can write

$$\overrightarrow{OA} = p\overrightarrow{a_1} + q\overrightarrow{a_2} + r\overrightarrow{a_3}$$

Where p, q and r are integers (since we define regular lattice points as those which can be arrived at using integral steps of the unit vectors corresponding to the lattice), and $\overrightarrow{a_1}, \overrightarrow{a_2}$, and $\overrightarrow{a_3}$ are the unit vectors corresponding to the lattice.

Let us define $\overrightarrow{S_0}$, and $\vec{S}$ as unit vectors along the direction of the incident beam and the direction of the diffracted beam respectively, as shown in Figure 28.2 above. If $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are perpendicular to $\overrightarrow{S_0}$ and $\vec{S}$ respectively, the path difference between the two rays is

$$\left|\overrightarrow{BA}\right| + \left|\overrightarrow{AC}\right|$$

From Figure 28.2, this is the same as

$$\left|\overrightarrow{EO}\right| + \left|\overrightarrow{OD}\right|$$

Since $\overrightarrow{S_0}.\overrightarrow{OA} = \left|\overrightarrow{OD}\right|$, and $-\vec{S}.\overrightarrow{OA} = \left|\overrightarrow{EO}\right|$,

The path difference is given by:

$$\left(\overrightarrow{S_0} - \vec{S}\right).\overrightarrow{OA}$$

Or, by convention, as

$$-\left(\vec{S} - \overrightarrow{S_0}\right).\overrightarrow{OA}$$

By definition, a path difference of λ equals a phase difference of 2π. Therefore the above path difference can be proportionally written as a phase difference given by:

$$-2\pi\frac{\left(\vec{S} - \overrightarrow{S_0}\right)}{\lambda}.\overrightarrow{OA}$$

Which we can rearrange and write as:

$$-2\pi\left(\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda}\right).\overrightarrow{OA}$$

We note that the vectors $\frac{\vec{S}}{\lambda}$, and $\frac{\overrightarrow{S_0}}{\lambda}$, both have L^{-1} dimensions, or reciprocal space dimensions. Therefore it can be written as $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$. However, it is important to note that at this time we have no information to suggest that it is a valid reciprocal lattice vector – meaning that it ends at valid reciprocal lattice point. It could perfectly well end in an empty location in reciprocal space, which does not correspond to a valid reciprocal lattice point. The condition that will indicate if a vector in reciprocal space is a valid reciprocal lattice vector is if it can be written as $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$ and if h, k, and l turn out to be integers. Therefore even though $\left(\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda}\right)$ can be written as $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$, in view of its dimensions being consistent with reciprocal space, at this time we have no information to suggest that h, k, and l are integers. Since the directions $\overrightarrow{S_0}$, and $\vec{S}$, and the wavelength λ, are all selections that we are free to make, we can easily pick values for these such that any of h, k, and l do not turn out to be integers.

The phase difference can be written as:

$$-2\pi\left(\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda}\right).\overrightarrow{OA} = -2\pi\left(h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}\right).\left(p\overrightarrow{a_1} + q\overrightarrow{a_2} + r\overrightarrow{a_3}\right)$$

$$= -2\pi(hp + kq + lr)$$

Where p, q and r are integers, while $h, k,$ and l need not be integers at this time.

However, for constructive interference between the two rays, the phase difference between them should be an integral multiple of 2π.

Therefore, for constructive interference, $(hp + kq + lr)$ should be an integer. Since p, q and r can change as we move to the next lattice point along $\overrightarrow{OA}$, and in principle, constructive interference should still occur since the orientation is the same, we require $hp + kq + lr$ to be an integer, regardless of the values p, q and r. This can only be accomplished with certainty if $h, k,$ and l are also integers. Therefore, the condition for constructive interference, and hence diffraction, requires that $h, k,$ and l are also integers.

In other words, we now find that diffraction requires $h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$ to be a valid reciprocal lattice vector – i.e. it ends at a valid reciprocal lattice point, or starts at the origin and ends at the valid reciprocal lattice point hkl, or can be designated as $\overrightarrow{H_{hkl}}$

Therefore the condition for diffraction in reciprocal space is that

$$\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} = \overrightarrow{H_{hkl}}$$

The left hand side of the equation contains information about the incident beam and the diffracted beam – both magnitude as well as direction, and the right hand side of the equation contains the information about the crystal lattice. This equation lays out the condition for diffraction in the context of reciprocal space.

We have therefore derived the condition for diffraction in real space as well as in reciprocal space, in this class. However, we have not stated anything specific on the origin of the waves that are getting diffracted. They could originate from a source external to the sample or even be the waves corresponding to the electrons within the sample itself.

In the upcoming classes, we will look at the interaction between electron waves that are due to the nearly free electrons within the sample, and the crystal structure of the same sample itself. In the very next class we will examine a pictorial manner in which the condition for diffraction in reciprocal space can be represented, and also examine how some common crystal structures get represented in reciprocal space.